AKTIVITAS MANDIRI - LARAVEL

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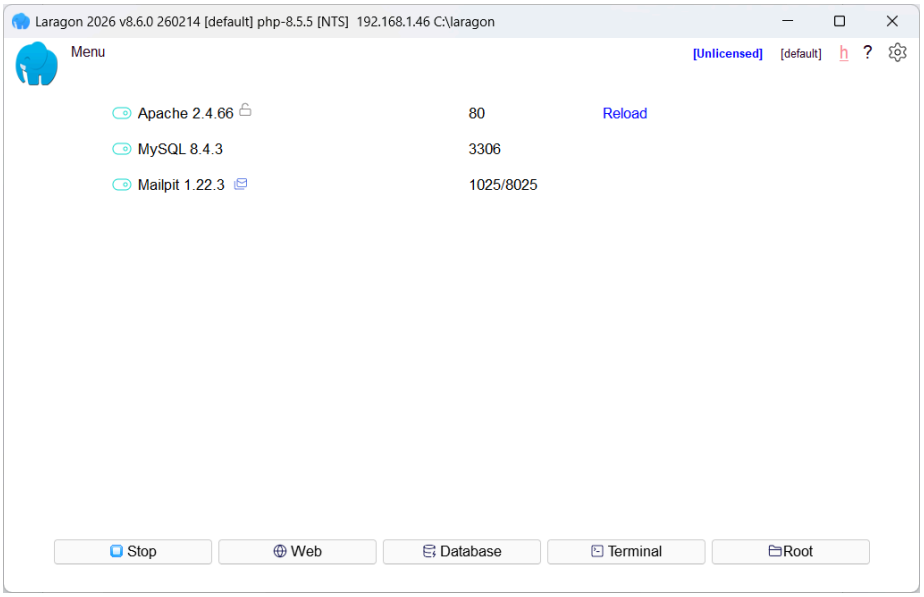
Kelas: PWEB A

AKTIVITAS MANDIRI 1 - Instalasi dan Setup Proyek Laravel

1. Instal XAMPP terbaru, pastikan PHP 8.2+ tersedia (cek: php --version di terminal)



Disini saya menggunakan laragon untuk opsi alternatif XAMPP dan versi php adalah sebagai berikut:

```
PS C:\laragon\www> php --version
PHP 8.5.5 (cli) (built: Apr  7 2026 19:24:35) (NTS Visual C++ 2022 x64)
Copyright (c) The PHP Group
Built by The PHP Group
Zend Engine v4.5.5, Copyright (c) Zend Technologies
    with Zend OPcache v8.5.5, Copyright (c), by Zend Technologies
```

2. Instal Composer dari getcomposer.org, verifikasi dengan: composer --version
   Di komputer saya, sudah menginstall composer. Jadi ini adalah versi dari composernya:

```
PS C:\laragon\www> composer --version
Composer version 2.9.7 2026-04-14 13:31:52
PHP version 8.5.5 (C:\laragon\bin\php\php-8.5.5-nts-Win32-vs17-x64\php.exe)
Run the "diagnose" command to get more detailed diagnostics output.
```

3. Buat proyek Laravel baru: composer create-project laravel/laravel siak-app
   Karena saya sudah menginstall laravel installer di global composer, jadi saya hanya perlu menjalankan *laravel new pitstop* (pitstop adalah nama web saya).

```
PS C:\laragon\www> laravel new pitstop

Creating a "laravel/laravel" project at "./pitstop"
Installing laravel/laravel (v13.3.0)
  - Installing laravel/laravel (v13.3.0): Extracting archive
Created project in C:\laragon\www\pitstop
Loading composer repositories with package information
Updating dependencies
Lock file operations: 106 installs, 0 updates, 0 removals
> @php -r "file_exists('.env') || copy('.env.example', '.env');"

  INFO  Application key set successfully.


 Which database will your application use? [SQLite]:
  [sqlite ] SQLite
  [mysql  ] MySQL
  [mariadb] MariaDB
  [pgsql  ] PostgreSQL (Missing PDO extension)
  [sqlsrv ] SQL Server (Missing PDO extension)
 > mysql
mysql


 Default database updated. Would you like to run the default database migrations? (yes/no) [yes]:
 >


  INFO  Nothing to migrate.

 Would you like to run npm install --ignore-scripts and npm run build? (yes/no) [yes]:
 >


added 63 packages, and audited 64 packages in 10s

16 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

> build
> vite build

vite v8.0.10 building client environment for production...
✓ 3 modules transformed.
computing gzip size...
public/build/manifest.json              0.33 kB │ gzip: 0.16 kB
public/build/assets/app-D05hqmaw.css   37.95 kB │ gzip: 9.38 kB
public/build/assets/app-34mOoJaZ.js     0.00 kB │ gzip: 0.02 kB

✓ built in 443ms

 Application ready in [pitstop]. You can start your local development using:

 → cd pitstop
 → composer run dev

 New to Laravel? Check out our documentation. Build something amazing!
```

4. Buat database baru siak_db di phpMyAdmin
   Membuat database dengan nama “pitstop” di phpMyAdmin

5. Konfigurasi file .env: isi DB_DATABASE, DB_USERNAME, DB_PASSWORD sesuai lokal

```
pitstop > .env
23  DB_CONNECTION=mysql
24  DB_HOST=127.0.0.1
25  DB_PORT=3306
26  DB_DATABASE=pitstop
27  DB_USERNAME=root
28  DB_PASSWORD=
29
```

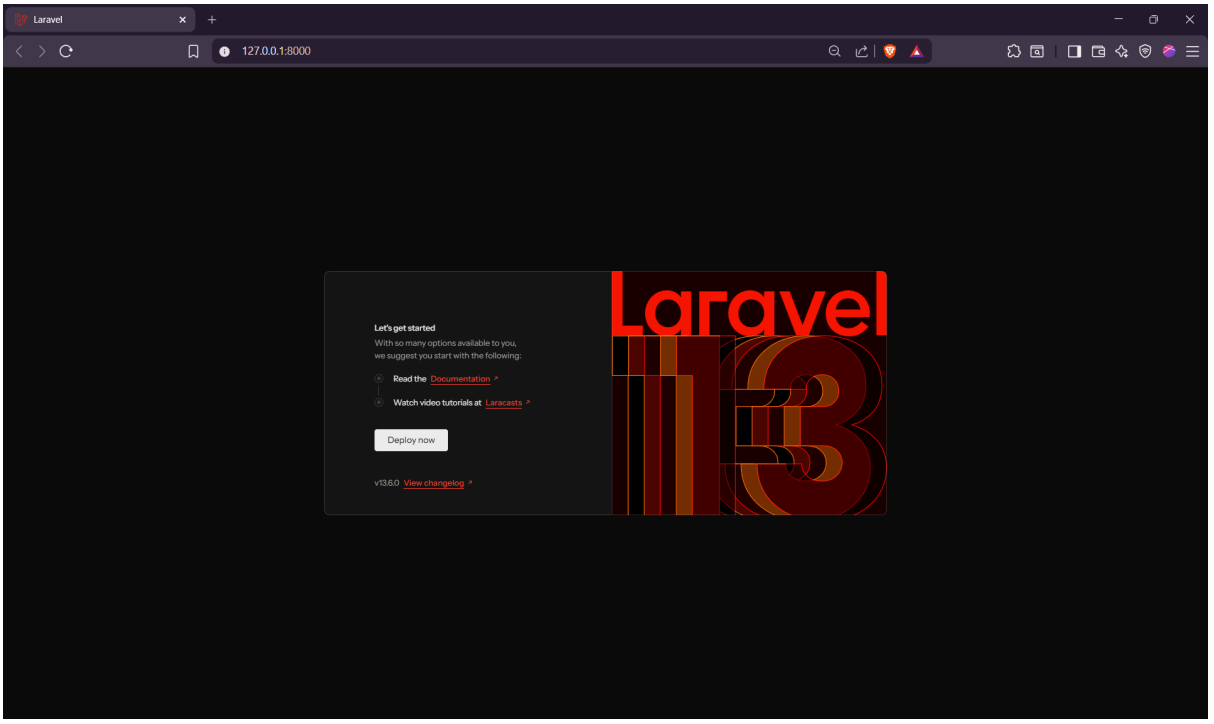
6. Jalankan: php artisan key:generate

```
PS C:\laragon\www\242410101063-PWEB> php artisan key:generate

  INFO  Application key set successfully.
```

7. Jalankan server: php artisan serve - Lalu buka http://127.0.0.1:8000 di browser

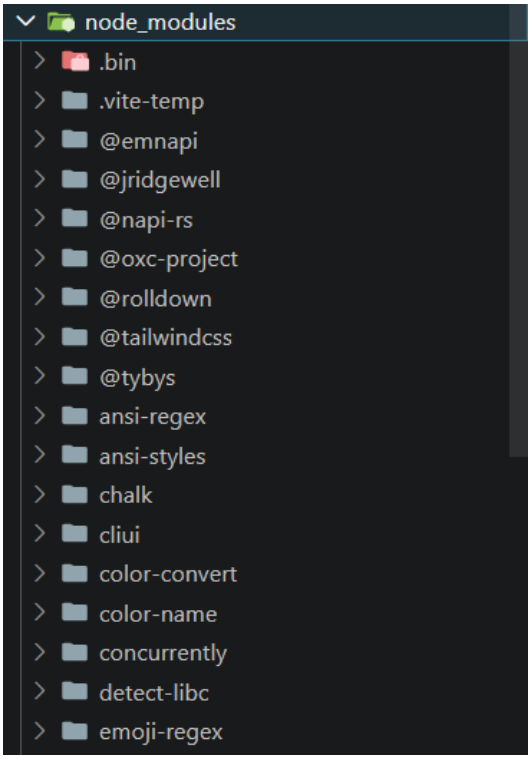
```
PS C:\laragon\www\242410101063-PWEB> php artisan serve

  INFO  Server running on [http://127.0.0.1:8000].

  Press Ctrl+C to stop the server
```

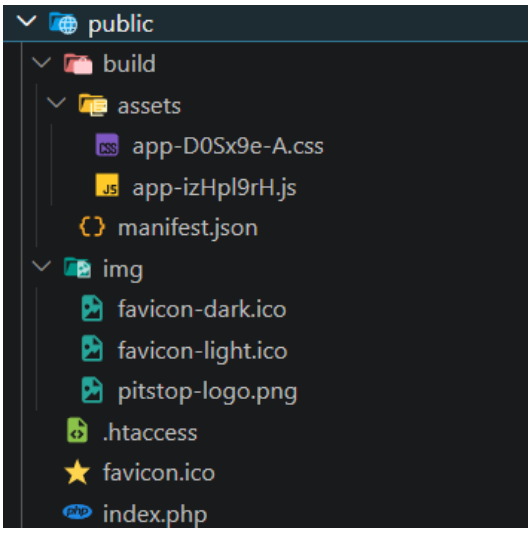
Lalu tampilan web awalnya:

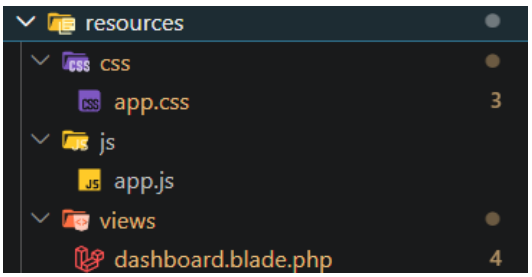


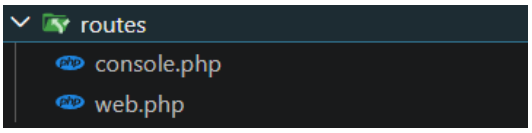
8. Eksplorasi struktur folder, buka setiap folder dan pahami fungsinya

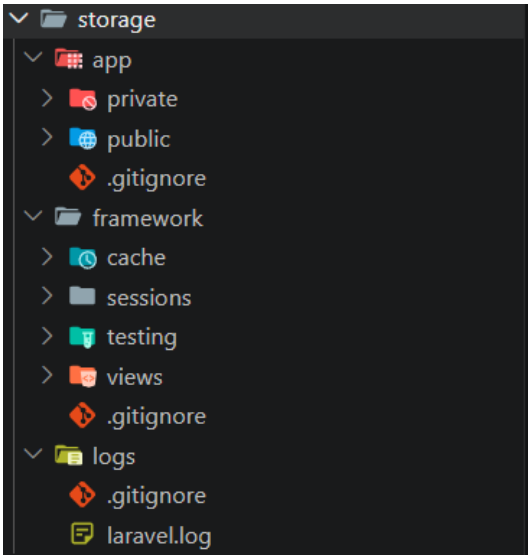
Folder	Fungsi
app Http\Controllers Controller.php DashboardController.php Models User.php Providers AppServiceProvider.php	
bootstrap cache .gitignore packages.php services.php app.php providers.php	
config app.php auth.php cache.php database.php filesystems.php logging.php mail.php queue.php services.php session.php	
database factories UserFactory.php migrations 0001_01_01_000000_create_users_tabl... 0001_01_01_000001_create_cache_tabl... 0001_01_01_000002_create_jobs_table... seeders DatabaseSeeder.php .gitignore	

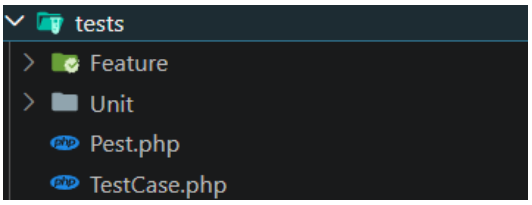

node_modules
.bin
.vite-temp
@emnapi
@jridgewell
@napi-rs
@oxc-project
@rolldown
@tailwindcss
@tybys
ansi-regex
ansi-styles
chalk
cliui
color-convert
color-name
concurrently
detect-libc
emoji-regex

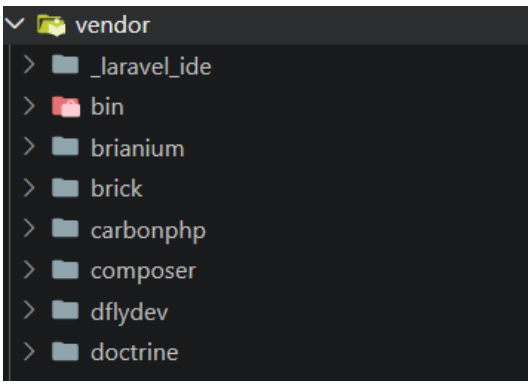

public
build
assets
app-D0Sx9e-A.css
app-izHpl9rH.js
manifest.json
img
favicon-dark.ico
favicon-light.ico
pitstop-logo.png
.htaccess
favicon.ico
index.php

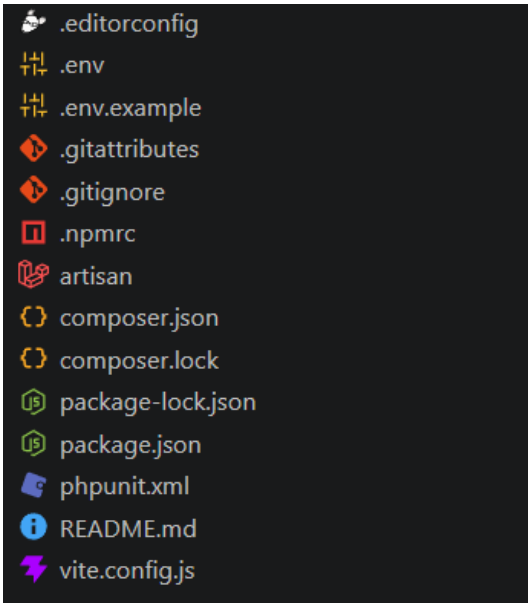

resources
css
app.css
3
js
app.js
views
dashboard.blade.php
4


routes
console.php
web.php


storage
app
private
public
.gitignore
framework
cache
sessions
testing
views
.gitignore
logs
.gitignore
laravel.log


tests
Feature
Unit
Pest.php
TestCase.php


vendor
_laravel_ide
bin
brianium
brick
carbonphp
composer
dflydev
doctrine



9. Jalankan: php artisan route:list

```
PS C:\laragon\www\242410101063-PWEB> php artisan route:list

  GET|HEAD  / ..................................................................... routes/web.php:5
  GET|HEAD  storage/{path} storage.local › vendor/laravel/framework/src/Illuminate/Filesystem/Filesy…
  PUT       storage/{path} storage.local.upload › vendor/laravel/framework/src/Illuminate/Filesystem…
  GET|HEAD  up vendor/laravel/framework/src/Illuminate/Foundation/Configuration/ApplicationBuilder.p…

                                                                                  Showing [4] routes
```

Ini adalah router awal dari laravel.

Aktivitas Mandiri 2 - Membuat Routing & Controller Dashboard

1. Buat Controller DashboardController:
   php artisan make:controller DashboardController

```
PS C:\laragon\www\242410101063-PWEB\pitstop> php artisan make:controller DashboardController

  INFO  Controller [C:\laragon\www\242410101063-PWEB\pitstop\app\Http\Controllers\DashboardController.php]
 created successfully.
```

2. Tambahkan method index() yang mengembalikan view('dashboard') di DashboardController

```
2 references | 0 implementations
class DashboardController extends Controller
{
    1 reference | 0 overrides
    public function index()
    {
        return view('dashboard');
    }
}
```

3. Daftarkan route di routes/web.php:
   Route::get('/dashboard', [DashboardController::class, 'index'])->name('dashboard');

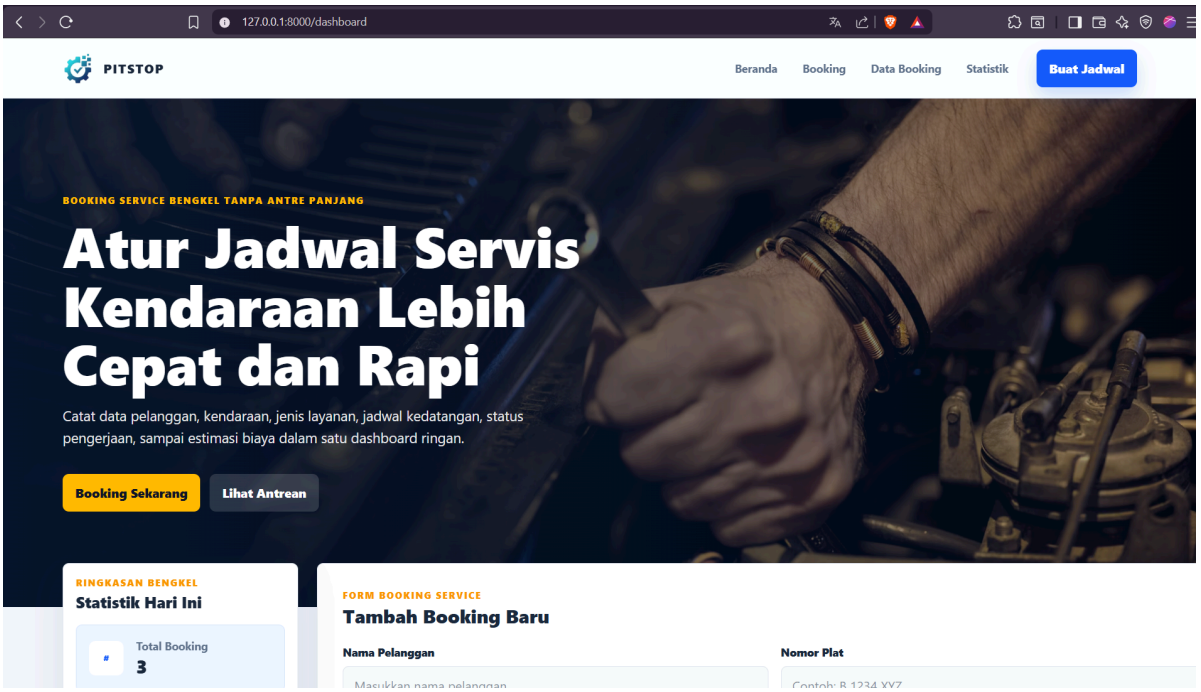
```php
<?php

use App\Http\Controllers\DashboardController;
use Illuminate\Support\Facades\Route;

Route::get('/', function () {
    return view('index');
});

Route::get('/dashboard', [DashboardController::class, 'index'])->name('dashboard');
```

4. Buat file view sementara resources/views/dashboard.blade.php berisi: <h1>Dashboard SIAK</h1>

```
resources > views > dashboard.blade.php > ...
<!doctype html>
<html lang="id">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>PitStop - Booking Service Bengkel</title>
    <link rel="icon" href="{{ asset('img/favicon-dark.ico') }}" media="(prefers-color-scheme: dark)">
    <link rel="icon" href="{{ asset('img/favicon-light.ico') }}" media="(prefers-color-scheme: light)">
    @vite(['resources/css/app.css', 'resources/js/app.js'])
</head>

<body class="min-h-screen overflow-x-hidden bg-[#f3f6fb] text-slate-800 antialiased">
    <header
        class="sticky top-0 z-20 flex flex-wrap items-center justify-between gap-4 border-b border-slate-200/90 bg-white/95 px-
        <a class="inline-flex items-center gap-3 font-black tracking-[0.08em]" href="#">
            <img src="{{ asset('img/pitstop-logo.png') }}" alt="Logo PitStop" class="h-10 w-10">
            <span>PITSTOP</span>
        </a>
```

Disini saya membuat dashboard yang sesuai dengan lanjutan projek kemarin.

5. Akses http://127.0.0.1:8000/dashboard di browser — pastikan tampil



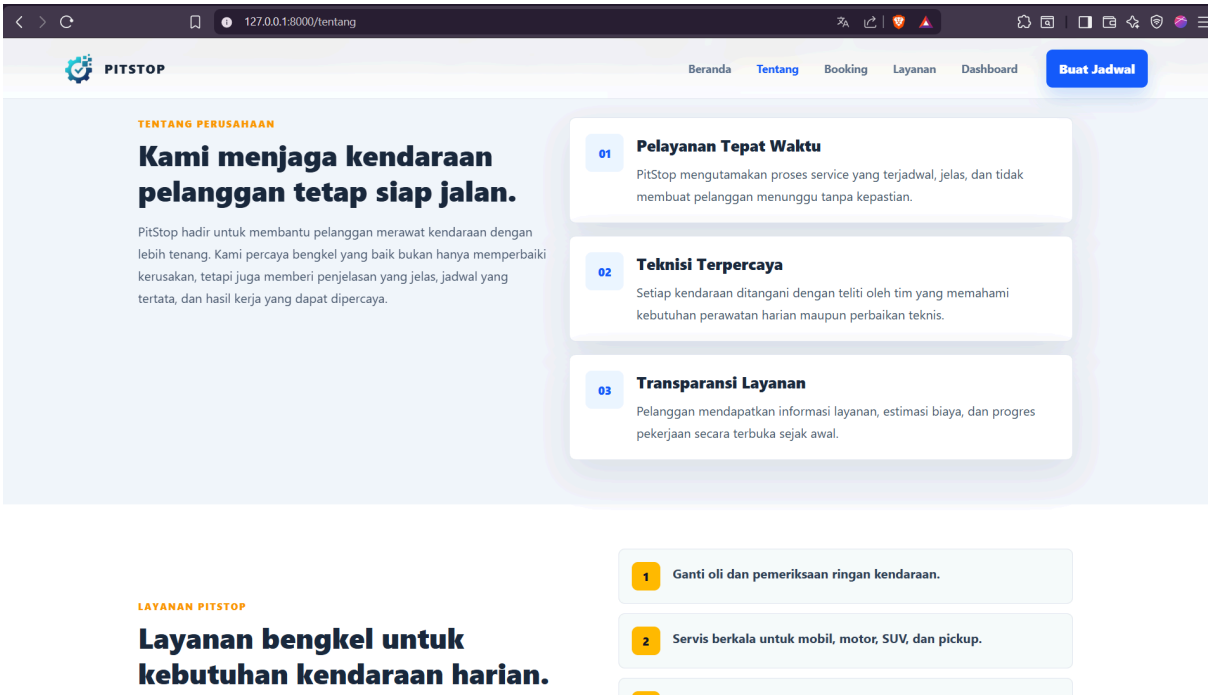
Di route ini sudah bisa tertampilkan.

6. Buat route /tentang dengan: Route::view('/tentang', 'tentang'); dan buat view-nya

```php
Route::view('/tentang', 'tentang');
```

```
resources > views > tentang.blade.php > ...
<!doctype html>
<html lang="id">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Tentang Kami - PitStop</title>
    <link rel="icon" href="{{ asset('img/favicon-dark.ico') }}" media="(prefers-color-scheme: dark)">
    <link rel="icon" href="{{ asset('img/favicon-light.ico') }}" media="(prefers-color-scheme: light)">
    @vite(['resources/css/app.css'])
</head>

<body class="min-h-screen overflow-x-hidden bg-[#f3f6fb] text-slate-800 antialiased">
    <header
        class="sticky top-0 z-20 flex flex-wrap items-center justify-between gap-4 border-b border-slate-200/90 bg-white/95 px-
        <a class="inline-flex items-center gap-3 font-black tracking-[0.08em]" href="{{ url('/') }}">
            <img src="{{ asset('img/pitstop-logo.png') }}" alt="Logo PitStop" class="h-10 w-10">
            <span>PITSTOP</span>
        </a>

        <input type="checkbox" id="menu-toggle" class="peer sr-only">
        <label for="menu-toggle"
            class="grid h-11 w-11 cursor-pointer place-items-center rounded-lg border border-slate-200 text-2xl leading-none te
            aria-label="Buka menu">
            &#9776;
        </label>

        <nav
            class="hidden w-full border-t border-slate-200 bg-white pt-4 shadow-[0_18px_45px_rgba(14,43,82,0.10)] peer-checked:
            <ul class="grid gap-4 text-sm font-bold text-slate-500 lg:flex lg:items-center lg:gap-8">
                <li><a class="transition hover:text-blue-600" href="{{ url('/') }}">Beranda</a></li>
                <li><a class="text-blue-600" href="{{ url('/tentang') }}">Tentang</a></li>
```

potongan kode view lanjutan dari projek kemarin.



7. Jalankan: php artisan route:list dan amati semua route yang terdaftar

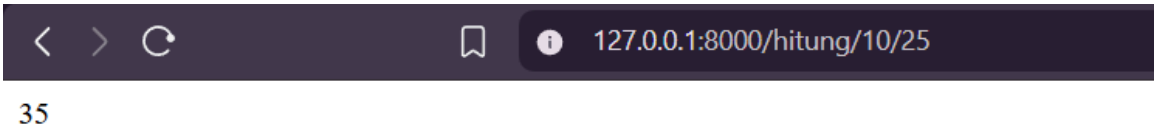
```
PS C:\laragon\www\242410101063-PWEB\pitstop> php artisan route:list

  GET|HEAD  / ..................................................... routes/web.php:6
  GET|HEAD  dashboard ............................... dashboard › DashboardController@index
  GET|HEAD  storage/{path} storage.local › vendor/laravel/framework/src/Illuminate/Filesys…
  PUT       storage/{path} storage.local.upload › vendor/laravel/framework/src/Illuminate/…
  GET|HEAD  tentang ...............................................................
  GET|HEAD  up vendor/laravel/framework/src/Illuminate/Foundation/Configuration/Applicatio…

                                                                  Showing [6] routes
```

Disini route-route tadi (dashboard, tentang) sudah terdaftar.

8. Coba buat route: Route::get('/hitung/{a}/{b}', fn($a, $b) => $a + $b); lalu akses /hitung/10/25
   Pada proyek saya sebenarnya tidak ada route yang seperti ini, jadi route ini hanya saya buat untuk percobaan dan tidak berhubungan dengan aplikasi utama.
   Ini routenya:

```
Route::get('/hitung/{a}/{b}', fn ($a, $b) => $a + $b);
```

   Lalu, ini jika kita akses route ini:



   Jadi di halaman web, ini akan menjalankan fungsi yang terdaftar di route. Tujuan dari aktivitas ini kemungkinan untuk memahami bagaimana cara menangani (handle) request GET dari URL yang telah terdaftar di dalam route.